Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **How the kit builds its sweep.** The ADALM-PHASER carries $N = 8$ elements spaced $d = 14$ mm apart. At 10.3 GHz the wavelength is 29.1 mm, so $kd = 173.2°$ of phase per unit of sin θ. In the Rectangular plot the kit holds the HB100 still and steps the element-to-element phase one $2.8125°$ LSB per point.

   (a) Find the change in $\sin\theta_0$ for one step, and the step in angle near broadside.

   $$\Delta(\sin\theta_0) = \frac{2.8125°}{173.2°} = 0.0162$$
   $$\Delta\theta_0 \approx 0.0162 \text{ rad} = 0.93°$$

   $\Delta\theta_0$ = $0.93°$

   (b) Find the same step in angle near $30°$.

   A fixed step in $\sin\theta$ is a larger step in θ where $\cos(\theta)$ is small:

   $$\Delta\theta_0 \approx \frac{0.93°}{\cos(30°)} = 1.07°$$

   $\Delta\theta_0$ = $1.07°$

   (c) Every point of the sweep comes from a different beam. In two or three sentences, explain why the trace still looks like the pattern of a single beam.

   The array factor depends only on $\psi = kd(\sin\theta_{\text{src}} - \sin\theta_0)$, and its magnitude is the same for ψ and $-\psi$. Stepping θ_0 past a fixed source runs ψ through the same values as moving the source past a beam held at a fixed θ_0, so both draw the same curve. The sweep's x-axis is the commanded angle θ_0, not a measured arrival angle.

   (d) State one thing the sweep cannot show that a measurement with one beam held still can, and give the reason.

   The pattern of each patch element. The sweep holds the source in one direction, so each patch receives the same level at every point, and that constant drops out of a trace read in dBc. A held beam with a moving source carries the source across each patch's pattern, which falls toward $\pm 90°$.

2. **The walk.** With Lab Preset 7 the kit holds the beam at 0° and plots the peak level against time while you carry the HB100 from -90° to $+90^\circ$ on a 1.2 m arc in 25 s. The beam's half-power width is 13.3°.

   (a) At a constant pace, how long does the source spend inside the half-power width of the main lobe?

   $t =$ 1.85 s

   $$\frac{180^\circ}{25\ \text{s}} = 7.2^\circ/\text{s}$$
   $$t = \frac{13.3^\circ}{7.2^\circ/\text{s}} = 1.85\ \text{s}$$

   (b) You walk through the main lobe at 70% of your average pace, then convert time to angle at the average pace. What half-power width does the trace report?

   $\theta_{\text{HP}} =$ 19.0°

   The source takes $1/0.7$ times as long to cross the lobe, so the lobe spans $1/0.7$ times as much time:

   $$\theta_{\text{HP}} = \frac{13.3^\circ}{0.7} = 19.0^\circ$$

   (c) Does the slow pace in part (b) change the level of the main lobe or of the first sidelobes on the trace? Explain in one or two sentences.

   No. The level at each instant is the pattern in the direction the source occupies at that instant, whatever its speed, so a slow pace stretches a lobe in time without changing its height.

   (d) Your arm drifts 10 cm outward, to 1.3 m, as you walk. By how much does the received level change?

   $\Delta L =$ -0.7 dB

   Received power falls as $1/r^2$:

   $$20\log_{10}\left(\frac{1.3\ \text{m}}{1.2\ \text{m}}\right) = 0.7\ \text{dB}$$

3. **Comparing the two traces.** A team records a sweep with the HB100 at boresight and a walk with the beam held at $0°$, both at 1.2 m. The sweep reads its first sidelobes at -12.6 and -13.4 dBc. The walk reads them at -11.6 and -14.7 dBc. Each reading carries about ± 1 dB from room reflections.

   (a) Do the two methods agree on the first sidelobes? Give the reason, using the uncertainty of a dBc value.

   A dBc value is the difference of two readings, each good to about ± 1 dB , so it is good to about ± 2 dB . Every one of the four values lies within 2 dB of the array factor's -12.8 dB , so the two methods agree. The 3.1 dB spread between the walk's two sidelobes is about what reflections in a classroom produce.

   (b) The team reads the walk's main-lobe width as $17°$ and the sweep's as $13°$. Which width do you report, and why?

   The sweep's. Its x-axis is the commanded angle, good to about one $0.93°$ step, while the walk's x-axis is time, which becomes angle only at a constant pace. A slow pace through the main lobe widens it on the walk's trace, as in Question 2(b).

   (c) Beyond $\pm 50°$ the sweep's lobes read about -17 dBc and the walk's about -21 dBc. Name two reasons the far lobes differ.

   The sweep has a floor about 23 dB below its peak, which adds to sidelobes that sit only a few decibels above it. The walk carries the HB100 across each patch's pattern, which falls toward $\pm 90°$ because the ground plane blocks the back; the sweep holds the HB100 in one direction, so that pattern does not shape its trace. Reflections also matter most where the pattern is weak.

   (d) The walk shows a lobe at -20 dBc near $+75°$ that is not in the sweep, and it moves when a classmate steps aside. What is it?

   A reflection off the classmate. A feature of the antenna stays put when people move; a feature that moves with them comes from the room.

4. **The beam at 30°.** The team now holds the beam at 30° for the walk, and places the HB100 at 30° for the sweep. For this array $\lambda/(Nd) = 0.260$ at 10.3 GHz.

   (a) Find the angles of the two first nulls of the 30° beam.

   The pattern slides in $\sin\theta$ (Lesson 18):

   $$\sin\theta_{\text{null}} = 0.5 \pm 0.260 = 0.240,\ 0.760$$
   $$\theta_{\text{null}} = 13.9^\circ,\ 49.5^\circ$$

   θ_{null} = $13.9^\circ,\ 49.5^\circ$

   (b) Explain why the main lobe reaches farther above its peak than below it, and whether the sweep and the walk should both show this.

   The nulls are an equal step of 0.260 apart in $\sin\theta$, and a step in $\sin\theta$ spans more angle where $\cos(\theta)$ is small, so the lobe reaches 19.5° above its peak and 16.1° below. Both methods run ψ through the same values, so both show it.

   (c) The array factor puts both first sidelobes of this beam at -12.8 dBc. The walk reads the outer one, near 60°, at about -15 dBc, while the sweep reads it near -12 dBc. Give the cause, and say whether one walk in a classroom can confirm it.

   The walk carries the HB100 out to 60°, where each patch receives about 3 dB less than at broadside; the sweep keeps the HB100 at 30° for every point, so the patch pattern does not change its shape. One walk cannot confirm it: the room alone can make two first sidelobes differ by about 3 dB, as large as the effect. Repeated turns of the kit on its stand, at a fixed distance, are the better test.

5. **Source distance.** Lesson 18 takes the course array's aperture as $D = Nd = 112$ mm. You place the HB100 1.2 m from the array face.

   (a) Compute the far-field distance for this array at 10.3 GHz ($\lambda = 29.1$ mm), and state whether the lab range qualifies.

   $r =$ 0.86 m

   $$r \geqslant \frac{2D^2}{\lambda} = \frac{2\,(0.112\text{ m})^2}{0.0291\text{ m}} = 0.86\text{ m}$$

   The 1.2 m range is past this, so the edge-to-center phase error of the incoming wavefront stays under the $22.5°$ the criterion allows.

   (b) You need to measure a larger array on the same 1.2 m range at 10.3 GHz. Find the largest aperture dimension that still satisfies the far-field criterion.

   $D =$ 132 mm

   Solve the criterion for D:

   $$D = \sqrt{\frac{r\lambda}{2}} = \sqrt{\frac{(1.2\text{ m})\,(0.0291\text{ m})}{2}} = 0.132\text{ m}$$

   so an aperture larger than about 132 mm needs a longer range.

   (c) Name two changes to the setup, not the antenna, that make the far sidelobes of either trace more trustworthy.

   Reduce the reflections: put absorber on the bench and nearby surfaces, keep people and equipment out of the path, and set the array and the HB100 at the same height away from large flat surfaces. Then repeat the measurement with the whole setup moved; a feature that moves with the room comes from a reflection, and one that stays with the array comes from the antenna.

Documentation: _____